DATA STRUCTURES AND ALGORITHMS

PROGRAM OUTPUTS

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Section: S9

VS Code-style editor and terminal output pages

Experiment 1 — Unit I

Basic Array Operations (Insertion, Deletion, Traversal)

ex1.c — Visual Studio Code

ex1.c

```c
#include <stdio.h>
int main() {
    int a[20], n, pos, value, i;
    printf("Enter array size: ");
    scanf("%d", &n);
    /* Array insertion, deletion and traversal */
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\DSA_Lab> gcc ex1.c -o output
PS C:\DSA_Lab> .\output

Enter array size: 5
Enter array elements: 2 4 6 8 0
Enter position and value to insert: 2 10
After Insertion: 2 10 4 6 8 0
Enter position to delete: 3
After Deletion: 2 10 6 8 0
Array Elements: 2 10 6 8 0
```

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Experiment 2 — Unit I

Dynamic Employee Record System using Structure Pointers

ex2.c — Visual Studio Code

ex2.c

```c
#include <stdio.h>
#include <stdlib.h>
struct Employee {
    int id;
    char name[30];
    float salary;
};
int main() {
    /* Read and display employee records */
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\DSA_Lab> gcc ex2.c -o output
PS C:\DSA_Lab> .\output

Enter number of employees: 2
Enter ID, Name and Salary: 101 Ravi 25000
Enter ID, Name and Salary: 102 Priya 30000

Employee Records
ID: 101 Name: Ravi Salary: 25000.00
ID: 102 Name: Priya Salary: 30000.00
```

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Experiment 3 — Unit II

Stack and Expression Conversion — Infix to Postfix

ex3.c — Visual Studio Code

ex3.c

```c
#include <stdio.h>
#include <ctype.h>
char stack[100];
int top = -1;
int precedence(char op) {
    /* Return operator precedence */
    return 0;
}
int main() {
    /* Convert infix expression to postfix */
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\DSA_Lab> gcc ex3.c -o output
PS C:\DSA_Lab> .\output

Enter infix expression: A+B*(C-D)
Postfix expression: ABCD-*+
```

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Experiment 4 — Unit II

Postfix Expression Evaluation using Stack

ex4.c — Visual Studio Code

ex4.c

```c
#include <stdio.h>
int stack[100], top = -1;
int main() {
    char postfix[100];
    /* Evaluate postfix expression using a stack */
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\DSA_Lab> gcc ex4.c -o output
PS C:\DSA_Lab> .\output

Enter postfix expression: 23*54*+
Result = 26
```

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Experiment 5 — Unit II

Simple Queue using Array

ex5.c — Visual Studio Code

ex5.c

```c
#include <stdio.h>
#define SIZE 100
int queue[SIZE];
int front = -1, rear = -1;
int main() {
    /* Insert, delete and display queue elements */
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\DSA_Lab> gcc ex5.c -o output
PS C:\DSA_Lab> .\output

Queue: 10 20 30
Deleted: 10
Queue: 20 30
```

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Experiment 6 — Unit II

Priority Queue using Unordered Array

ex6.c — Visual Studio Code

ex6.c

```c
#include <stdio.h>
struct Item {
    int value, priority;
};
int main() {
    /* Insert items and delete highest-priority item */
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\DSA_Lab> gcc ex6.c -o output
PS C:\DSA_Lab> .\output

Priority Queue: (10,p2) (20,p5) (30,p1)
Deleted: 20 (priority 5)
Priority Queue: (10,p2) (30,p1)
```

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Experiment 7 — Unit III

Singly Linked List (Insertion, Deletion, Traversal)

ex7.c — Visual Studio Code

ex7.c

```c
#include <stdio.h>
#include <stdlib.h>
struct Node {
    int data;
    struct Node *next;
};
int main() {
    /* Perform linked-list operations */
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\DSA_Lab> gcc ex7.c -o output
PS C:\DSA_Lab> .\output

List: 10 -> 20 -> 30 -> NULL
List: 20 -> 30 -> NULL
```

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Experiment 8 — Unit III

Circular Queue using Linked List

ex8.c — Visual Studio Code

ex8.c

```c
#include <stdio.h>
#include <stdlib.h>
struct Node {
    int data;
    struct Node *next;
};
int main() {
    /* Implement circular queue operations */
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\DSA_Lab> gcc ex8.c -o output
PS C:\DSA_Lab> .\output

Circular Queue: 10 20 30
Deleted: 10
Circular Queue: 20 30
```

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Experiment 9 — Unit IV

Binary Search Tree Traversals

ex9.c — Visual Studio Code

ex9.c

```c
#include <stdio.h>
#include <stdlib.h>
struct Node {
    int data;
    struct Node *left, *right;
};
int main() {
    /* Display inorder, preorder and postorder */
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\DSA_Lab> gcc ex9.c -o output
PS C:\DSA_Lab> .\output

In-order: 20 30 40 50 60 70 80
Pre-order: 50 30 20 40 70 60 80
Post-order: 20 40 30 60 80 70 50
```

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Experiment 10 — Unit V

Graph Search Algorithms (BFS and DFS)

ex10.c — Visual Studio Code

ex10.c

```c
#include <stdio.h>
#define V 5
int main() {
    int graph[V][V];
    /* Traverse graph using BFS and DFS */
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\DSA_Lab> gcc ex10.c -o output
PS C:\DSA_Lab> .\output

BFS: 0 1 2 3 4
DFS: 0 1 3 4 2
```

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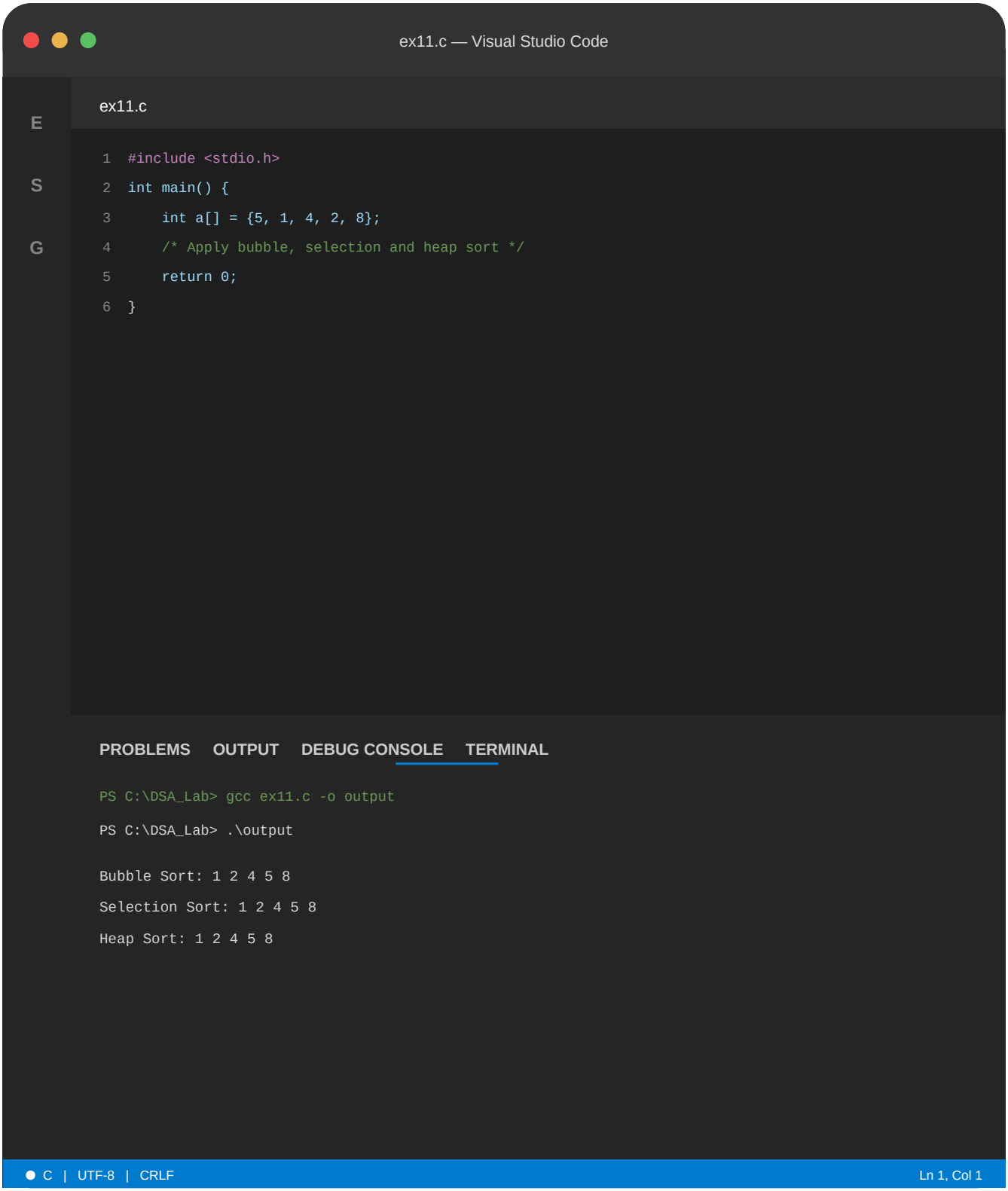
Experiment 11 — Unit V

Bubble, Selection and Heap Sorts



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Experiment 12 — Unit V

Hashing with Collision Resolution using Linear Probing

ex12.c — Visual Studio Code

ex12.c

```c
#include <stdio.h>
#define SIZE 10
int main() {
    int hash[SIZE];
    /* Insert keys using linear probing */
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
PS C:\DSA_Lab> gcc ex12.c -o output
PS C:\DSA_Lab> .\output

Hash Table
0 : -1
1 : -1
2 : -1
3 : 23
4 : 43
5 : 13
6 : -1
7 : 27
8 : -1
```

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